

FULLSTACK APPLICATION FUNDAMENTALS

Object-Oriented Programming

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Outline

- 1 Layered Architecture
- 2 Data Layer
- 3 Backend Layer
- 4 FrontEnd Layer
- 5 Computing Resources

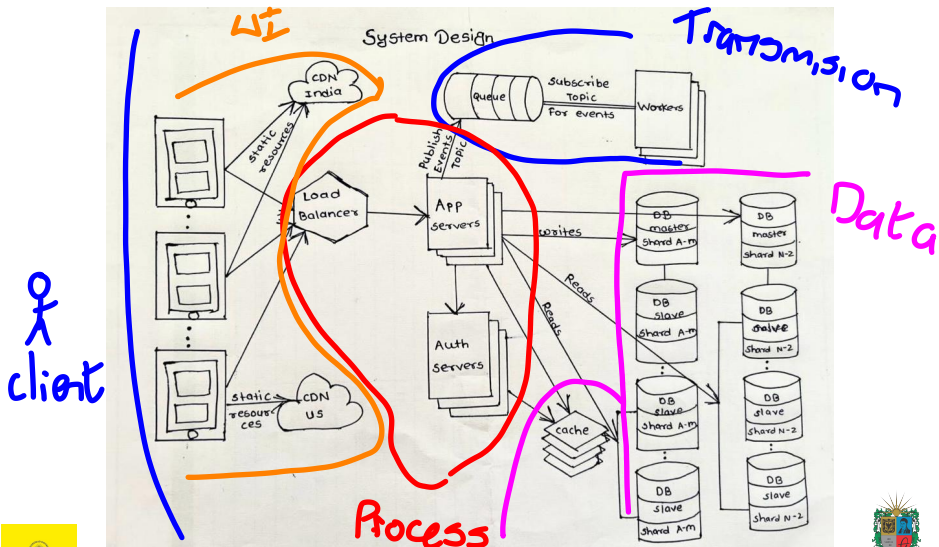


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Systems Design applied to Software Architectures



What is a System Architecture?

- A **system architecture** is the **structure** of a system that *defines* its **components**, **interactions**, and **relationships**. $\Rightarrow$ O.O.P.
- A **system architecture** is the **blueprint** of a system that *guides* its **development** and **implementation**. $\hookrightarrow$ *template + recipe*
- A **system architecture** is the foundation of a **system** that *ensures* that it **meets** the **needs** of its **users**. $\hookrightarrow$ *solve the problem*



Monolithic System Architecture

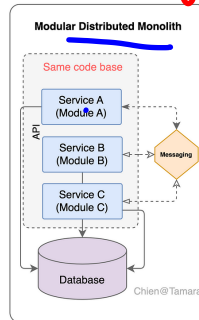
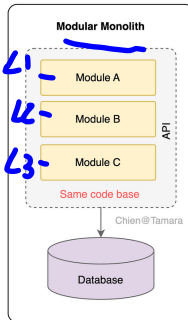
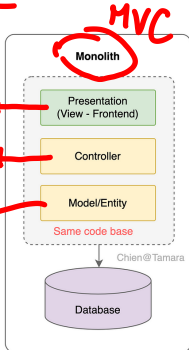
- A **monolithic system architecture** is a **single-tier architecture** that consists of a **single unit** that performs all the functions of the system.
 → 1 project → 1 machine
- It is **simple**, **easy to develop**, and **maintain**, but it is **not scalable** and flexible. It is **typically used** for **small systems** that do not require high performance or reliability.
 → my first project

each Folder

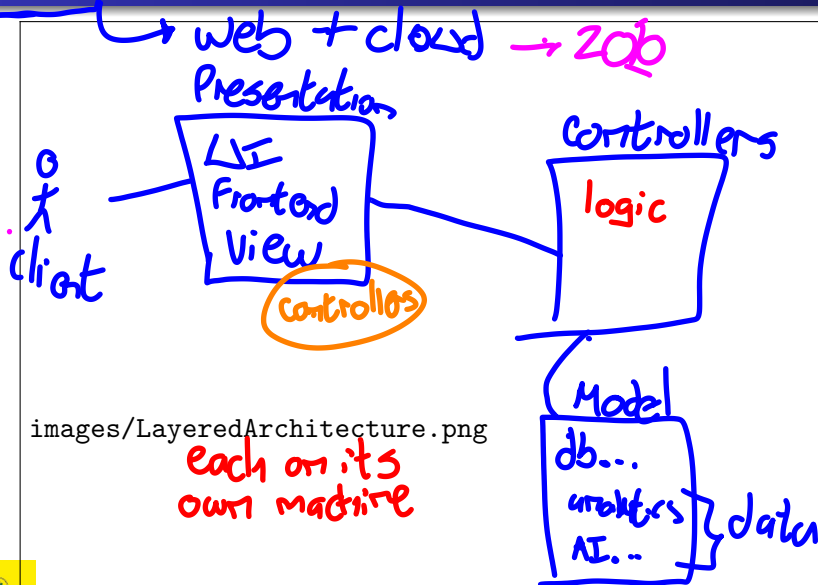
LI

Logic

Data Access



Layered Architecture Pattern



images/LayeredArchitecture.png

each on its
own machine



Packages in Java

Packages are a way of structuring the Java **namespace** using dotted package names. → **Folder structure**

- **Creating Packages:** To create a package, you just have to create a directory with a package-info.java file.
- **Importing Packages:** To import a package, you can use the import statement.
- **Third-party Packages:** Java has a set of third-party packages that you can use in your projects.
- **Maven:** Maven is a build automation tool used primarily for Java projects.

Demo time!

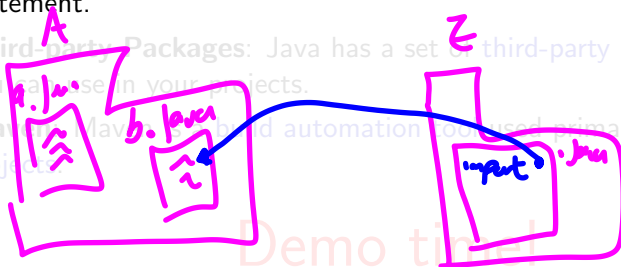


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↳ util

Math

• Spring Boot

Demo time!



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project management
packages
Demo time!



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Data System Concepts

Key Points of Data Systems:

- Data modeling is the process of designing the structure and organization of data.
- Data storage is the process of storing data in a structured or unstructured format.
- Data retrieval is the process of accessing and retrieving data from a storage system.
- Data transformation is the process of modifying and transforming data.
- Data security is the process of protecting data from unauthorized access and ensuring its integrity and confidentiality.

what

Format

sources
transform

?



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 - resources
expensive
 - read
extract
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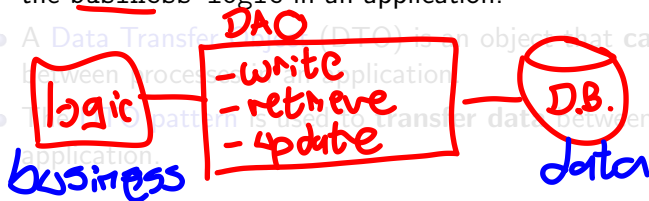
no damage



Data Access Objects and Data Transfer Objects

Data Access Objects (DAOs) and Data Transfer Objects (DTOs) are design patterns used to separate the data access logic from the business logic in an application.

- A Data Access Object (DAO) is an object that provides an **abstract interface** to some type of database or other persistence mechanism.
- The DAO pattern is used to **separate the** data access logic from the business logic in an application.



Demo time!



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- A **Data Transfer Object** (DTO) is an object that **carries data** between processes in an application.
- The **DTO pattern** is used to **transfer data** between processes in an application.

only attributes

→ database only

List of Objects

Demo time!

List of rows

name	age	place	address
.		.	



Objects Persistence

- **Serialization:** It is the process of converting an object into a stream of bytes. *over the net → fast, flyweight*
- **Deserialization:** It is the process of converting a stream of bytes into an object.

Demo time!

- **JSON:** It is a lightweight data-interchange format that is easy for *humans* to read and write and easy for *machines* to parse and generate.

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Backend Concepts

Key Points of Backend Systems:

- A **backend system** is a software system that provides the **logic** and functionality to support the front-end of an application.
- A **backend system** typically consists of a server, a database, and an application server.
- A **server** is a computer that provides **services** to other computers over a network.
- An **application server** is a software framework that provides an environment for **running web applications**.
- A **database** is a collection of data that is **organized and stored** in a defined format.



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Connection with Data Layer

- The **backend layer** is responsible for managing the **data layer** and providing the **logic** and **functionality** to *support the front-end* of an application.
- The **connection** between the backend and data layers is typically managed through an **application programming interface (API)**.
- An **API** is a set of **rules** and **protocols** that allows different software applications to **communicate** with each other.
- The **API** provides a way for the front-end of an application to **interact** with the backend and access the data stored in the database.
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FrontEnd Layer

- The **front-end** is the **client-side** of the application and everything that the user interacts with.
- The **front-end** is the **presentation layer** of the application.
- The **front-end** is the **user interface** and the **user experience**.



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Model-View-Controller (MVC) Pattern

- The **Model-View-Controller** (MVC) is a software architectural pattern that separates the application into three main components: **model**, **controller**, and **view**.
- The **model** is responsible for managing the **data** and **business** logic of the application.
- The **view** is responsible for **displaying** the data to the user and providing a way for the user to interact with the application.
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Java Swing & Java FX

- **Java Desktop Applications** are applications that **run on a user's computer** and provide a graphical user interface (GUI) for the user to interact with.
- **Java Swing** is a set of **GUI components** that can be used to create **desktop applications** in Java.
- **JavaFX** is a set of **GUI components** that can be used to create **desktop applications** in Java.
- **JavaFX** is the successor to **Java Swing** and provides a more modern, **flexible** way to create **desktop applications**.
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- **JavaFX Components** are the building blocks of a **JavaFX user interface**.
- **JavaFX Components** are organized into a **scene graph**, which is a hierarchical structure that represents the **layout** and **organization** of the user interface.
- **JavaFX Components** can be **customized** and **styled** using CSS.
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- **Memory Profiling:** It is the process of analyzing the **memory usage** of a program.
- **Memory Leaks:** A common problem in programming where memory is allocated but **never deallocated**.
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Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/object-oriented-programming>

